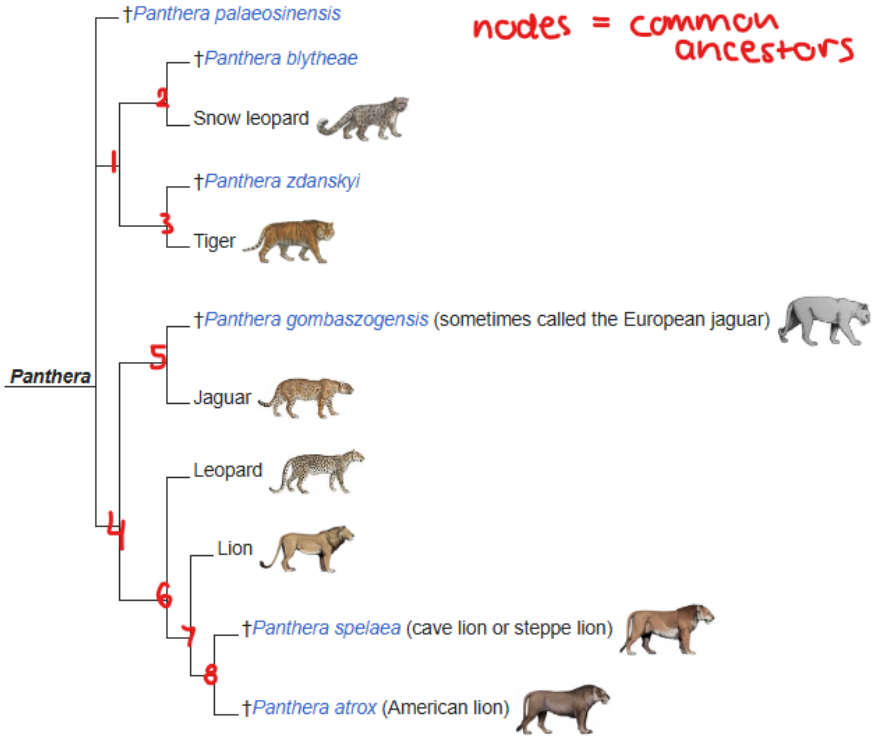
LAB 08 ANSWERS REPORT

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COMP3742, TONSLEY
COMPLETED: 11/10/2026

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1. **Define a Prolog knowledge base for the Genus Panthera. Refer to the Panthera cladogram on the previous page and the Wikipedia site https://en.wikipedia.org/wiki/Panthera .**




```
parent(panthera, palaeosinensis)
parent(panthera, node1)

parent(node1, node2)
parent(node1, node3)

parent(node2, blytheae)
parent(node2, snow_leopard)

parent(node3, zdanskyi)
parent(node3, tiger)

parent(node4, node5)
parent(node4, node6)

parent(node5, gombaszogensis)
parent(node5, jaguar)

parent(node6, leopard)
parent(node6, node7)

parent(node7, lion)
parent(node7, node8)

parent(node8, spelaea)
parent(node8, atrox)
```

2. **Define Prolog rules and facts to describe the Panthera cladogram in the form of a Prolog Knowledge base.**

Question: isn't this the same as the previous question? Does the knowledge base describe all rules and facts here or is there a difference?

```
%%% RULES %%%

% Immediate ancestors
immediate_ancestors(X,Y) :-
        parent(A, X),
        parent(A, Y),
        X \- Y.

% Direct descendant
direct_descendant(X, Y) :-
        parent(Y,X).

% Indirect descendant
descendant(X, Y) :-
    parent(Z, X),
    descendant(Z, Y).
```

3. **Query the node two generations prior (grandparent node) of "Cave Lion". Ensure to show the query and results.**

```
10 ?- parent(X, Y), parent(Y, spelaea).
X = node7,
Y = node8
```

4. **Find the direct descendants (child nodes) of "Ancestor 4". Ensure to show the query and results.**

```
11 ?- parent(node4, X).
X = node5 ;
X = node6.
```

5. **Query the immediate ancestral node (parent node) of 'Lion". Ensure to show the query and results.**

```
12 ?- parent(X, lion).
X = node7.
```